

Statistical mechanics of the n -queens lattice gas: Monte Carlo simulations and thermodynamic integration

Zong-Yue Liu¹ and Lei Wang²

¹*Department of Physics, Nanjing University, Nanjing 210093, China*

²*Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China*

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We investigate the n -queens problem as a lattice gas—a model in which N queens are placed on an $N \times N$ chessboard with pairwise repulsive interactions along shared rows, columns, and diagonals—from the perspective of statistical mechanics. The ground states are exactly the $Q(N)$ solutions of the classical N -queens problem, whose entropy per queen $s_0 \approx \ln N - \gamma$ ($\gamma \approx 1.944$) exhibits a characteristic constraint hierarchy: each successive geometric constraint—columns, then diagonals—reduces the entropy from the free-placement value $\ln N$ by a definite constant. We derive the exact high-temperature energy $E/N \rightarrow 5/3$ as $N \rightarrow \infty$. Extensive Monte Carlo simulations with 10^8 sweeps per temperature point for $N = 8$ to 1024 reveal that the specific heat per queen C_v/N converges to a universal function of T as $N \rightarrow \infty$, with a non-divergent peak $C_v^{\max}/N \approx 1.63$ at $T^* \approx 0.235 J$, establishing the absence of a thermodynamic phase transition. Combined with the exactly known high-temperature entropy, the convergence of C_v/N enables a thermodynamic integration of C_v/T from $T = \infty$ to $T = 0$ that recovers the ground-state entropy—and hence the Simkin constant γ —purely from Monte Carlo data, providing the first **[LW:no need to claim first]** *physical* determination of this fundamental combinatorial constant. Thermodynamic integration yields $\gamma_{\text{MC}} = 1.946$ at $N = 1024$, within 0.1% of the precise combinatorial value.

I. INTRODUCTION

The N -queens problem—placing N mutually non-attacking queens on an $N \times N$ chessboard—is among the oldest combinatorial problems in mathematics, dating back to the work of Bezzel in 1848 [1]. Despite its long history, the asymptotic enumeration of solutions was resolved only recently. Simkin [2] proved that the number of solutions $Q(N)$ satisfies

$$Q(N) \sim \left(\frac{N}{e^\gamma}\right)^N, \quad (1)$$

building on the breakthrough of Bowtell and Keevash [3] and earlier bounds by Luria and Simkin [4]. Simkin's proof introduces *queenons*—limit objects defined as probability measures on the unit square—and characterizes γ as the solution of a convex optimization problem over this space; the upper bound employs the entropy method while the lower bound is established by a randomized construction algorithm, yielding $\gamma \in [1.939, 1.945]$. Nobel, Agrawal, and Boyd [5] subsequently refined these bounds to $\gamma = 1.944001$ (precision $\sim 10^{-7}$) by solving Simkin's convex programs with large-scale Newton methods. All of these determinations are purely combinatorial. Equation (1) yields a ground-state entropy per queen $s_0 = \lim_{N \rightarrow \infty} \ln Q(N)/N = \ln N - \gamma$.

The form of the ground-state entropy is suggestive. On an $N \times N$ lattice, a single particle placed freely in one row has $\ln N$ units of entropy—the maximal combinatorial freedom per particle. The N -queens entropy $s_0 = \ln N - \gamma$ can therefore be read as: each queen retains the full lattice entropy $\ln N$, reduced by a constant γ that encodes the cumulative cost of geometric constraints. This constant decomposes naturally through

a hierarchy of constraints with increasing stringency [6]:

- *Free placement* (row constraint only): $\Omega = N^N$, $s = \ln N$.
- *Permutation* (row + column): $\Omega = N!$, $s \approx \ln N - 1$.
- *N -queens* (row + column + two diagonals): $\Omega \approx (N/e^\gamma)^N$, $s \approx \ln N - \gamma$.

The column constraint costs exactly one unit of entropy (the Stirling correction); the two diagonal constraints together cost $\gamma - 1 \approx 0.94$ units. That each geometric constraint reduces the entropy by a well-defined constant, independent of N in the large- N limit, hints at a deeper statistical-mechanical structure underlying the combinatorics.

This **[LW:it is unclear what this "is" across the paragraph]** observation motivates the present work. We embed the N -queens problem in a lattice gas with tunable temperature [7, 8] and perform extensive Monte Carlo simulations. Throughout this paper we set $k_B = 1$, so that entropy is dimensionless and temperature has units of energy. A central finding is that the specific heat per queen, C_v/N , converges to a size-independent curve for sufficiently large N , exhibiting a Schottky-type peak but no phase transition. Combined with the exactly known high-temperature entropy $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N} \approx \ln N + 1$, this convergence enables a thermodynamic integration

$$s_0 = \frac{S(\infty)}{N} - \int_0^\infty \frac{C_v}{NT} dT,$$

which extracts the ground-state entropy—and hence the Simkin constant γ —purely from Monte Carlo simulation

data. This provides the first *physical* route to γ , complementary to the combinatorial and optimization-based approaches that have dominated to date [2]. Our thermodynamic extraction yields $\gamma_{\text{MC}} = 1.946$ at $N = 1024$ (within 0.1% of the precise value [5]), demonstrating that a single Monte Carlo campaign can determine a fundamental combinatorial constant through the laws of thermodynamics.

The paper is organized as follows. Section II defines the model. Section III presents analytical results: the ground-state entropy and constraint hierarchy, and the exact high-temperature limit. Section IV presents Monte Carlo results. Section V carries out the thermodynamic integration. Section VI concludes.

II. MODEL AND SIMULATION METHODOLOGY

We consider an $N \times N$ square lattice with open boundary conditions. Each site (i, j) ($i, j = 1, \dots, N$) carries an occupation variable $n_{ij} \in \{0, 1\}$. The Hamiltonian counts mutually attacking queen pairs:

$$H = J \sum_{\langle (i,j), (i',j') \rangle_{\text{attack}}} n_{ij} n_{i'j'}, \quad (2)$$

where the sum runs over all pairs sharing a row ($i = i'$), column ($j = j'$), main diagonal ($i - j = i' - j'$), or anti-diagonal ($i + j = i' + j'$), and $J > 0$. We set $J = 1$ throughout.

The lattice decomposes naturally into *attack lines*: N rows, N columns, $2N - 1$ main diagonals, and $2N - 1$ anti-diagonals. Writing n_α for the number of queens on line α , the Hamiltonian takes the compact form

$$H = \sum_{\alpha \in \mathcal{L}} \binom{n_\alpha}{2}, \quad (3)$$

where $\mathcal{L}$ denotes the set of all $6N - 2$ attack lines and $\binom{n}{2} = n(n-1)/2$ counts the number of attacking pairs on each line. This decomposition into line contributions is the starting point for the high-temperature calculation (Sec. III B).

We work in the canonical ensemble with fixed particle number N and use Kawasaki dynamics [9] (queen–vacancy exchange) **[LW:why this ? why not queen-queen exchange ?]** with the Metropolis acceptance criterion [10]. One *sweep* consists of N attempted exchanges. Statistical errors are estimated by jackknife resampling with 200 bins [11]. We simulate eight system sizes $N = 8, 16, 32, 64, 128, 256, 512, 1024$ on a uniform grid of 280 temperature points spanning $T = 0.05$ to $500 J$, with 10^8 measurement sweeps and 2×10^6 thermalization sweeps at every temperature point. The grid places 160 points in the specific-heat peak region ($T = 0.1025$ – $0.500 J$ in steps of $0.0025 J$), 11 points in the low-temperature region ($T = 0.050$ – $0.100 J$, step

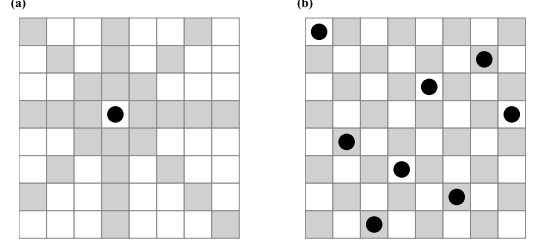


FIG. 1. (a) A single queen (black circle) on an 8×8 board. Gray squares are attacked along the row, column, and two diagonal directions; white squares are unattacked. (b) One of the $Q(8) = 92$ non-attacking ground-state configurations.

$0.005 J$), and the remaining 109 points at intermediate and high temperatures up to $T = 500 J$. All eight sizes share the identical temperature grid and measurement parameters, ensuring a self-consistent data set. The simulations were carried out using 260 CPU cores in parallel, with each core handling all eight system sizes at a single temperature point in serial; the total wall-clock time was approximately nine hours. We verified ergodicity by running independent simulations from multiple random initial configurations (seeds); all runs yielded statistically consistent results for E/N and C_v/N at every temperature, confirming that the Kawasaki dynamics explores the configuration space adequately within our thermalization and measurement windows. The schematic of the model and the relevant length scales are shown in Fig. 1.

III. ANALYTICAL RESULTS

A. Ground-state entropy and constraint hierarchy

For N queens, the ground states ($E = 0$) correspond to the $Q(N)$ non-attacking configurations. Using Eq. (1), the entropy per queen is

$$s_0 = \lim_{N \rightarrow \infty} \frac{\ln Q(N)}{N} = \ln N - \gamma, \quad \gamma \approx 1.94400. \quad (4)$$

The three-level constraint hierarchy (Table I) provides a physical interpretation. Starting from a fully unconstrained model where each queen independently chooses one of N columns ($\Omega_{\text{free}} = N^N$, $s_{\text{free}} = \ln N$), the column non-repetition constraint reduces the count to $N!$ permutations. By Stirling's approximation, $\ln N! = N \ln N - N + O(\ln N)$, so the entropy per queen is

$$s_{\text{perm}} = \frac{\ln N!}{N} = \ln N - 1 + O\left(\frac{\ln N}{N}\right). \quad (5)$$

Thus the column constraint costs exactly one unit of entropy—the Stirling correction arising from the indistinguishability of column assignments.

The two diagonal constraints further reduce the entropy to $s_0 = \ln N - \gamma$, costing an additional $\gamma - 1 \approx 0.94$ units, or approximately $(\gamma - 1)/2 \approx 0.47$ units per diagonal family. The hierarchy of costs is physically natural: the column constraint, being a global permutation on N equivalent lines, removes exactly one unit of freedom per queen, while each diagonal family—whose lines have unequal lengths ranging from 1 to N —removes roughly half a unit.

That each geometric constraint reduces the per-queen entropy by a constant independent of N suggests that the finite-temperature thermodynamics may also admit a simple analytical description. We turn to this question next, beginning with the high-temperature limit.

B. High-temperature limit

At $T \rightarrow \infty$, all $\binom{N^2}{N}$ configurations with N queens on the $N \times N$ board are equally probable. We derive the exact energy using linearity of expectation [8].

Writing the energy as a sum over attack lines [Eq. (3)], each line of length k contributes $\binom{n}{2} = \frac{1}{2} \sum_{i \neq j} n_i n_j$, where the sum runs over all pairs of sites on that line. By linearity of expectation,

$$\left\langle \binom{n}{2} \right\rangle = \binom{k}{2} P(n_i=1, n_j=1). \quad (6)$$

The probability that two given sites are simultaneously occupied is a combinatorial identity independent of their positions:

$$P(n_i=1, n_j=1) = \frac{\binom{N^2-2}{N-2}}{\binom{N^2}{N}} = \frac{N(N-1)}{N^2(N^2-1)}. \quad (7)$$

The total number of attacking site pairs $S_{\text{tot}} = \sum_{\alpha} \binom{k_{\alpha}}{2}$ is obtained by summing over all attack lines. Rows contribute $S_{\text{row}} = N \binom{N}{2} = N^2(N-1)/2$; columns contribute identically. Each diagonal family has $2N-1$ diagonals of lengths $k = N-|d|$ ($d = 0, \pm 1, \dots, \pm(N-1)$), giving

$$S_{\text{diag}} = \sum_{d=-(N-1)}^{N-1} \binom{N-|d|}{2} = \frac{N(N-1)(2N-1)}{6} \quad (8)$$

TABLE I. Entropy per queen under successive constraints for the N -queens problem. The column constraint costs exactly 1 unit; the two diagonal constraints cost $\gamma - 1 \approx 0.94$ units combined.

Model	Constraints	Ω	$s = \frac{1}{N} \ln \Omega$
Free	Row only	N^N	$\ln N$
Permutation	Row + column	$N!$	$\ln N - 1$
N -queens	Row + col + 2 diag	$\approx (N/e^{\gamma})^N$	$\ln N - \gamma$

per family. Summing all four families:

$$S_{\text{tot}} = 2 \cdot \frac{N^2(N-1)}{2} + 2 \cdot \frac{N(N-1)(2N-1)}{6} = \frac{N(N-1)(5N-1)}{3}. \quad (9)$$

Combining Eqs. (6)–(9), the exact per-queen energy at $T \rightarrow \infty$ is

$$\left. \frac{\langle E \rangle}{N} \right|_{T \rightarrow \infty} = \frac{S_{\text{tot}}}{N} \cdot \frac{N(N-1)}{N^2(N^2-1)} = \frac{(5N-1)(N-1)}{3N(N+1)} \xrightarrow{N \rightarrow \infty} \frac{5}{3}. \quad (10)$$

This result is exact for all finite N and is verified by Monte Carlo simulations (Fig. 3): at high T , E/N converges to the value given by Eq. (10), which approaches $5/3$ from below with corrections of order $O(1/N)$.

IV. FINITE-TEMPERATURE MONTE CARLO RESULTS

A. Convergence diagnostics

Before presenting thermodynamic observables, we verify that the Monte Carlo simulations have reached equilibrium and that the measurements are statistically independent. Figure 2 shows the acceptance rate and the integrated autocorrelation time τ_{int} of the energy as functions of temperature for several system sizes.

The acceptance rate [panel (a)] increases monotonically with T , reaching ~ 0.38 – 0.47 at $T = 1 J$; it drops below 10^{-5} for $T \lesssim 0.1 J$, reflecting the near-perfect freezing into non-attacking ground states. For $N \geq 32$ the curves collapse, confirming convergence to the thermodynamic limit.

The autocorrelation time [panel (b)] exhibits a sharp peak at $T \approx 0.055$ – $0.10 J$, well below the specific heat maximum $T^* \approx 0.235 J$, reaching $\tau_{\text{int}} \approx 5.0 \times 10^4$ sweeps for $N = 1024$. Even at this worst case, 10^8 measurement sweeps yield ~ 2000 independent samples—adequate for the essentially featureless low-temperature regime where both E/N and C_v/N approach zero exponentially. Crucially, at the physically important specific heat peak ($T \approx 0.235 J$), $\tau_{\text{int}} \approx 85$ – 200 sweeps for all N , providing more than 5×10^5 independent samples per temperature point and ensuring high statistical precision in the region that matters most. At $T/J > 0.4$, τ_{int} falls to ~ 5 – 10 sweeps for all system sizes. We additionally verified ergodicity by comparing independent runs from different random initial configurations, all yielding statistically consistent results.

B. Energy

Figure 3 shows E/N vs. T for $N = 8$ to 1024 . The curves for $N \geq 32$ are nearly indistinguishable, confirming rapid convergence to the thermodynamic limit. At high temperatures, E/N approaches the exact finite- N value [Eq. (10)], which itself converges to $5/3$ as $N \rightarrow \infty$.

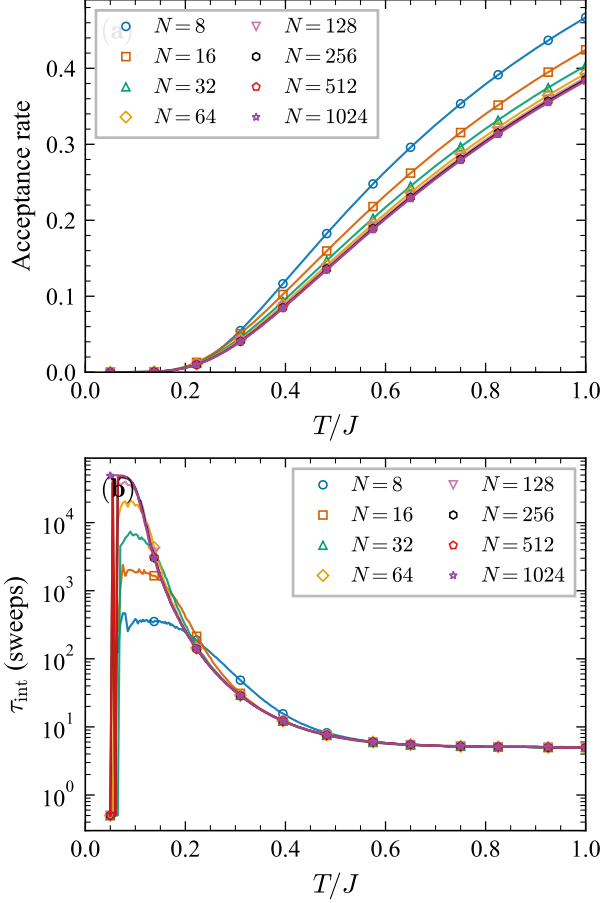


FIG. 2. Convergence diagnostics for the Monte Carlo simulations. (a) Acceptance rate of the Kawasaki (queen–vacancy exchange) [LW:consider making this log y scale] moves as a function of temperature for $N = 8$ to 1024. The rate drops below 10^{-5} for $T \lesssim 0.1 J$ and rises monotonically, reaching 0.38–0.47 at $T = 1 J$. (b) Integrated autocorrelation time τ_{int} of the energy in units of sweeps (log scale). The sharp peak at $T \approx 0.055$ – $0.10 J$ reflects low-temperature freezing and lies well below the specific heat maximum $T^* \approx 0.235 J$; at T^* itself, $\tau_{\text{int}} \approx 85$ – 200 sweeps, yielding more than 5×10^5 independent samples from 10^8 sweeps. [LW:legend (b) overlapped with curve]

C. Specific heat convergence

The specific heat per queen $C_v/N = (\langle E^2 \rangle - \langle E \rangle^2)/(T^2 N)$ is shown in Fig. 4. For $N \geq 32$, the C_v/N curves collapse onto a single function of T , indicating that the thermodynamic limit has been reached. Table II lists the peak values.

The key observation is that C_v^{max}/N saturates near 1.63 for $N \geq 32$ and does *not* diverge with system size. This rules out a thermodynamic phase transition and identifies the crossover as a Schottky-type anomaly [12]—a broad peak in C_v arising from thermal excitation across

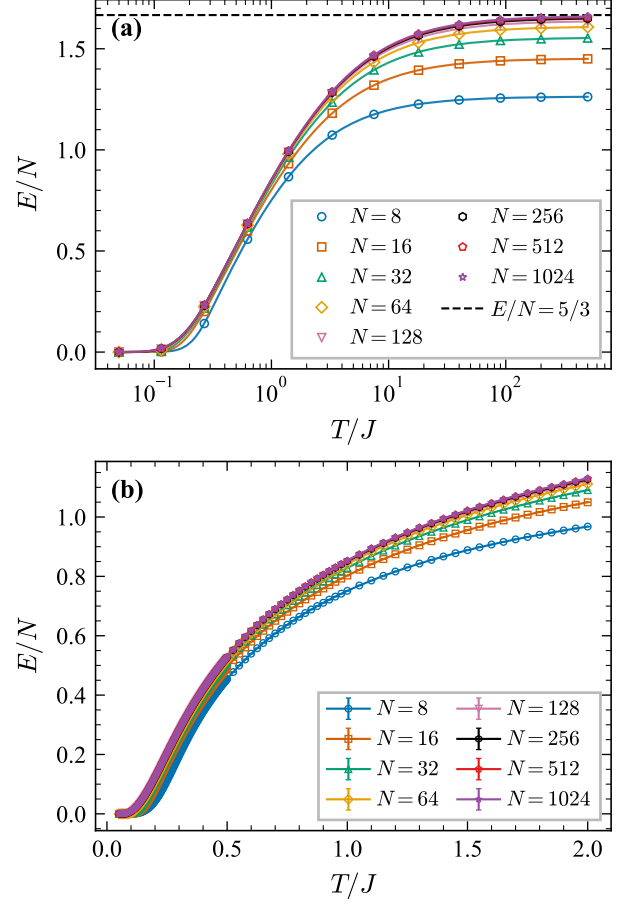


FIG. 3. Energy per queen E/N vs. temperature for $N = 8$ to 1024. (a) Logarithmic temperature scale over the full range $T = 0.05$ – $400 J$; the dashed line marks the analytical $N \rightarrow \infty$ high- T limit $E/N = 5/3$. (b) Linear scale, zoomed to $T \leq 2 J$, with error bars.

a finite energy gap, without any symmetry breaking. The peak temperature $T^* \approx 0.235 J$ stabilizes for $N \geq 64$, and the peak height converges monotonically from above: the residual finite-size correction between $N = 128$ and $N = 1024$ is only $\Delta C_v^{\text{max}}/N \approx 0.001$, well within the

TABLE II. C_v/N peak scaling from dedicated high-resolution simulations at $N = 8$ to 128 with dense temperature sampling in $T \in (0.20, 0.30) J$. The peak height saturates near 1.63 for $N \geq 32$, ruling out a divergent susceptibility and hence a continuous phase transition.

N	T_{peak}	C_v^{max}/N	error
8	0.281	1.6915	0.0002
16	0.227	1.6783	0.0006
32	0.231	1.6288	0.0005
64	0.237	1.6216	0.0004
100	0.237	1.6229	0.0004
128	0.237	1.6225	0.0004

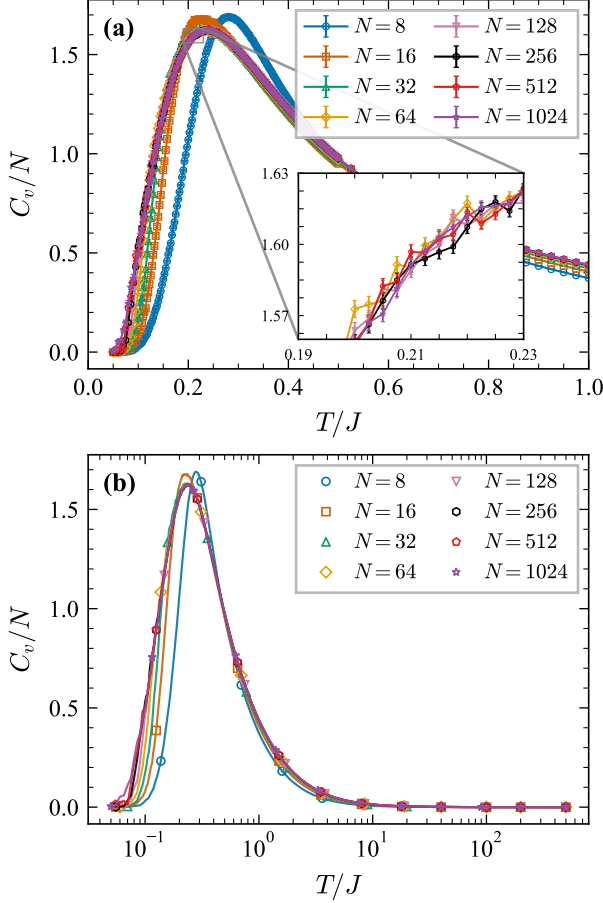


FIG. 4. Specific heat per queen C_v/N vs. T for $N = 8$ to 1024. (a) Peak region ($T \leq 1 J$, linear scale) with error bars; the inset enlarges the peak vicinity ($T = 0.19$ – $0.23 J$) for $N \geq 64$, revealing subtle size-dependent shifts in the peak height and position. (b) Full temperature range on a logarithmic T scale. The curves for $N \geq 32$ collapse onto a single function, confirming convergence to the thermodynamic limit.

statistical error bars.

The convergence of C_v/N to a definite limiting function has a powerful consequence: since both the high-temperature entropy $S(\infty)/N$ and the ground-state entropy s_0 are independently known, the integral $\int_0^\infty C_v/(NT) dT$ must equal their difference. We exploit this in the next section.

V. THERMODYNAMIC INTEGRATION AND ENTROPY SELF-CONSISTENCY

The convergence of C_v/N to a universal function enables a powerful thermodynamic self-consistency check. We compute the entropy difference between $T = \infty$ and $T = 0$ in two independent ways and verify that they agree.

Combinatorial route.—At $T \rightarrow \infty$, all configurations are equally probable, so the entropy per queen is

$$\frac{S(\infty)}{N} = \frac{1}{N} \ln \binom{N^2}{N}. \quad (11)$$

For $N \ll N^2$, applying the Stirling approximation $\ln n! = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O(n^{-1})$:

$$\begin{aligned} \ln \binom{N^2}{N} &= \ln(N^2!) - \ln N! - \ln(N^2 - N)! \\ &= N \ln N - (N^2 - N) \ln(1 - 1/N) + \Delta_S, \end{aligned} \quad (12)$$

where the first two terms come from the $n \ln n - n$ part of Stirling's formula and the correction $\Delta_S = \frac{1}{2} \ln[N^2/(2\pi N(N^2 - N))] = -\frac{1}{2} \ln(2\pi(N - 1))$ collects the subleading logarithms. Expanding $\ln(1 - 1/N) = -1/N - 1/(2N^2) - \dots$, the leading terms give $-(N^2 - N) \ln(1 - 1/N) = N - \frac{1}{2} + O(N^{-1})$, so

$$\frac{S(\infty)}{N} = \ln N + 1 + O\left(\frac{\ln N}{N}\right). \quad (13)$$

At $T = 0$, only ground states contribute: $S(0)/N = \ln Q(N)/N \approx \ln N - \gamma$.

Thermodynamic route.—The fundamental relation

$$S(\infty) - S(0) = \int_0^\infty \frac{C_v}{T} dT \quad (14)$$

relates the entropy difference to the specific heat. Dividing by N and substituting the results above:

$$\int_0^\infty \frac{C_v}{NT} dT = \frac{S(\infty) - S(0)}{N} = 1 + \gamma + O\left(\frac{\ln N}{N}\right). \quad (15)$$

This is a remarkable relation: the left side involves $C_v(T)$ measured by simulation over the entire temperature range, while the right side involves only the *combinatorial* constants 1 (from the Stirling correction) and γ (from the Simkin asymptotic formula).

Rearranging Eq. (15), we can extract the ground-state entropy per queen directly from the Monte Carlo data:

$$s_0^{\text{MC}} = \frac{S(\infty)}{N} - \int_0^\infty \frac{C_v}{NT} dT, \quad (16)$$

where $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$ is exact. This provides a purely thermodynamic determination of the ground-state degeneracy without direct enumeration.

The numerical integration uses 280 temperature grid points spanning from $T = 0.05 J$ to $400 J$; the low- T cutoff contributes negligibly since C_v/T is exponentially suppressed below $T \approx 0.1 J$, and the high- T cutoff at $400 J$ is sufficient because $C_v/T \propto T^{-3}$ decays rapidly, contributing less than 0.01% of the integral beyond this point. The trapezoidal rule on the logarithmic grid introduces discretization errors well below the statistical uncertainty of the Monte Carlo data.

TABLE III. Ground-state entropy from thermodynamic integration. $s_0^{\text{MC}} = S(\infty)/N - \int_0^\infty (C_v/NT) dT$, where $S(\infty)/N = \frac{1}{N} \ln \binom{N^2}{N}$ is exact. For $N \leq 16$, s_0^{MC} is compared with the exact ground-state entropy $s_0^{\text{exact}} = \ln Q(N)/N$. For $N \geq 32$, we extract $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$ and compare with the precise value $\gamma = 1.944001$ [5]. All data use a 280-point temperature grid with 10^8 measurement sweeps per point.

N	s_0^{MC}	s_0^{exact}	γ_{MC}	Deviation
8	0.566	0.565	—	0.1%
16	1.031	1.032	—	0.1%
32	1.633	—	1.833	5.7%
64	2.274	—	1.885	3.0%
128	2.943	—	1.909	1.8%
256	3.620	—	1.925	1.0%
512	4.308	—	1.931	0.69%
1024	4.985	—	1.946	0.11%

Table III summarizes the results. For $N = 8$ and $N = 16$, where exact values of $Q(N)$ are known ($Q(8) = 92$, $Q(16) = 14\,772\,512$), s_0^{MC} agrees with the exact $s_0^{\text{exact}} = \ln Q(N)/N$ to within 0.1%, validating the simulation and integration procedure.

For $N \geq 32$, where exact $Q(N)$ is unknown, we use s_0^{MC} to extract the Simkin constant via $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$. The extracted values converge toward the known asymptotic value $\gamma = 1.944001$ [5]: $\gamma_{\text{MC}} = 1.83$ ($N = 32$), 1.89 ($N = 64$), 1.91 ($N = 128$), 1.93 ($N = 256$), 1.93 ($N = 512$), and 1.946 ($N = 1024$), with the deviation decreasing from 5.7% to 0.11% at $N = 1024$. The dominant source of error at moderate N is the finite-size correction to the Simkin formula, which makes $\gamma_{\text{eff}}(N) = \ln N - s_0(N)$ systematically smaller than the asymptotic γ ; the MC integration itself introduces only sub-percent errors, as confirmed by the $N = 8$ and $N = 16$ benchmarks. The monotonic convergence of γ_{MC} toward $\gamma = 1.944001$ with increasing N —reaching 0.11% accuracy at $N = 1024$ —confirms the mutual consistency of the Monte Carlo thermodynamics, the Stirling approximation for $S(\infty)$, and the Simkin combinatorial formula.

VI. CONCLUSIONS

We have presented a statistical-mechanical investigation of the n -queens lattice gas [LW:call the problem **n-queens lattice gas**] through extensive Monte Carlo simulations. Our main findings are:

1. The ground-state entropy $s_0 = \ln N - \gamma$ decomposes through a constraint hierarchy: the column constraint costs exactly 1 unit of entropy per queen (the Stirling correction), while the two diagonal

constraints together cost $\gamma - 1 \approx 0.94$ units. That each constraint reduces the per-queen entropy by an N -independent constant suggests a statistical-mechanical structure amenable to analytical treatment.

2. At $T \rightarrow \infty$, the exact energy per queen is $E/N = (5N-1)(N-1)/[3N(N+1)] \rightarrow 5/3$, derived from a linearity-of-expectation argument applied to the uniform random placement of queens.
3. The specific heat per queen $C_v/N(T)$ converges to a universal function of T as $N \rightarrow \infty$. The peak height ≈ 1.63 does not diverge, ruling out a thermodynamic phase transition and identifying the crossover as a Schottky anomaly with effective gap $\Delta = J$.
4. Thermodynamic integration of C_v/T recovers the ground-state entropy per queen $s_0^{\text{MC}} = S(\infty)/N - \int (C_v/NT) dT$, agreeing with exact values at $N = 8$ and 16 to within 0.1%. For $N \geq 32$, the extracted Simkin constant $\gamma_{\text{MC}} = \ln N - s_0^{\text{MC}}$ converges to $\gamma = 1.944001$ [5], reaching 1.946 at $N = 1024$ (0.11% deviation). This nontrivial self-consistency check connects the Simkin combinatorial formula, the Stirling approximation for the high-temperature entropy, and the Monte Carlo specific heat data across the entire temperature range.

These results establish the queen lattice gas as a model system in which the interplay between combinatorial and thermodynamic quantities can be exploited quantitatively: a single Monte Carlo campaign, combined with exactly known high-temperature entropy, suffices to determine a fundamental combinatorial constant through the laws of thermodynamics.

Extending to still larger system sizes ($N \gtrsim 1024$) would further sharpen the thermodynamic determination of γ , potentially bringing the Monte Carlo estimate to sub-0.1% accuracy and rivaling direct combinatorial methods.

VII. ACKNOWLEDGMENTS

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